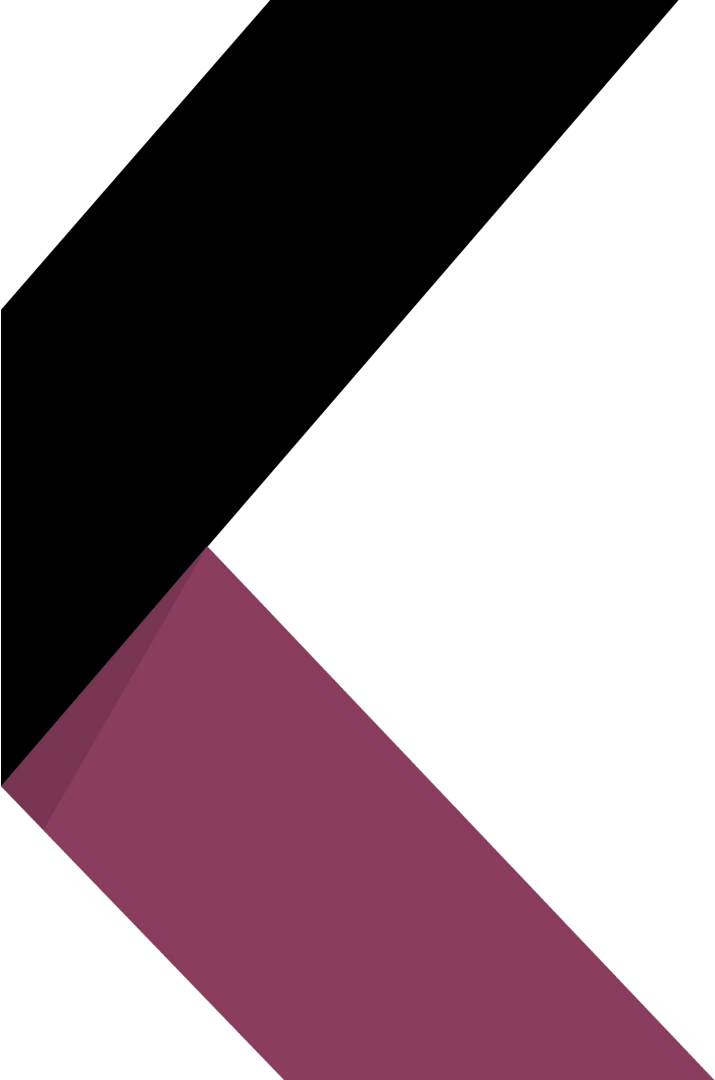


Permanent Magnet Brushless Machine

A decorative graphic on the right side of the slide. It consists of a large black triangle pointing downwards, which is partially overlapped by a smaller, semi-transparent maroon triangle pointing upwards. The intersection of these two triangles creates a darker maroon shape.



Introduction

Permanent Magnet Brushless Machine - Overview

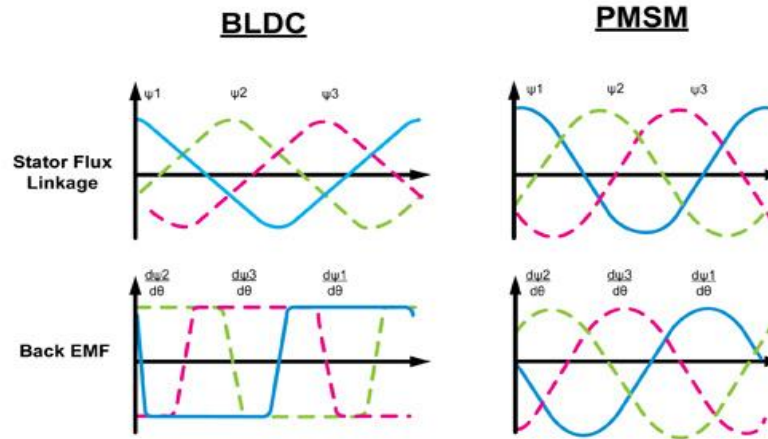
- Among various types of motor drives, permanent magnet (PM) brushless motor drives, especially the PM synchronous motor drive, are currently the most attractive motor drives for electric vehicle (EV) propulsion.
- Their key features, namely **high power density** and **high efficiency**, are attributed to the use of high-energy PM material. Actually, they are becoming dominant in the market share of EV motor drives. However, these PM brushless motor drives still have some shortcomings such as **high cost** and **thermal instability** of the PM material.
- The family of PM brushless machines has two main members: the **PM synchronous machine** and the **PM BLDC machine**. The PM synchronous machine has been widely used for EV propulsion; meanwhile, the PM BLDC machine is becoming attractive.



PM Brushless – Classification

Brushless synchronous motors can be categorized as follows according to **the shape of the back emf** developed at the air gap:

- **Trapezoidal**, the machine is powered by CSI inverter, BLDC (Brushless DC) machine;
- **Sinusoidal**, the machine is powered by VSI inverters with sinusoidal voltages (PMSM).



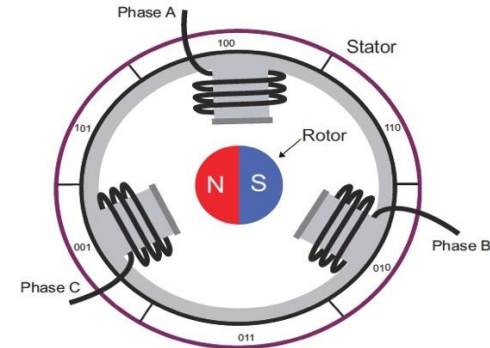
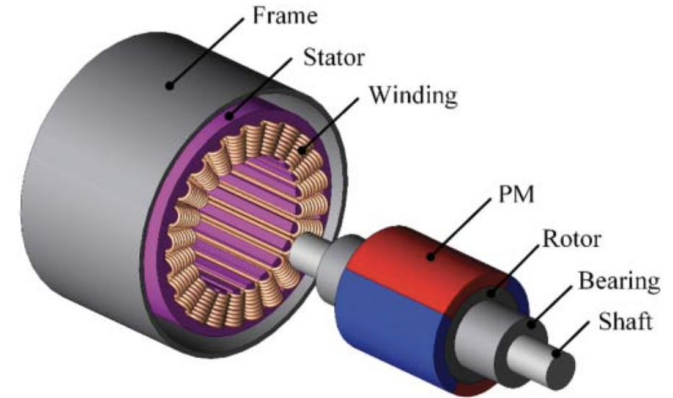


PM Brushless – Structure

The basic structure of the PM brushless machine, including both the PMSM and BLDC types, consists of:

- a stator housing a **winding**, typically three-phase connected in a star, through which the motor itself is powered;
- a rotor equipped with **permanent magnets** capable of establishing a magnetic field in the air gap quite similar to that produced by the excitation winding placed on the rotor of normal synchronous motors.

Compared with the induction machine, it is relatively simpler because of the **absence of cage bars** and end-rings.



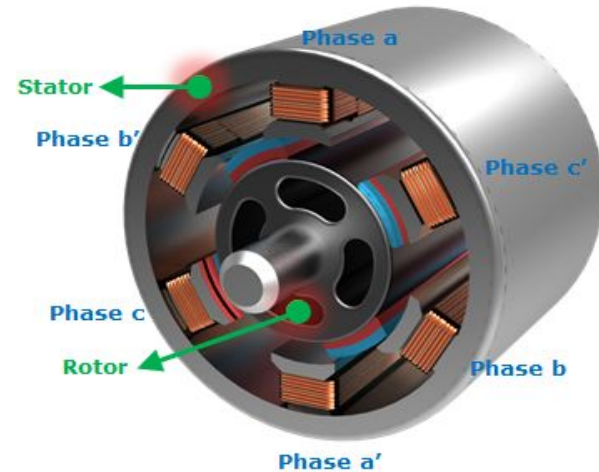


PMSM – Working principle

In the machine, the rotor rotates **at the same speed** as the rotating magnetic field of the stator, so the mechanical speed of the rotor coincides with the synchronous speed.

$$n_s = \frac{60f}{p}$$

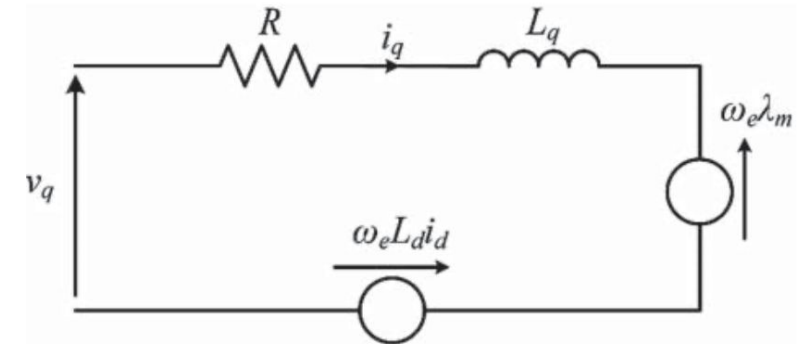
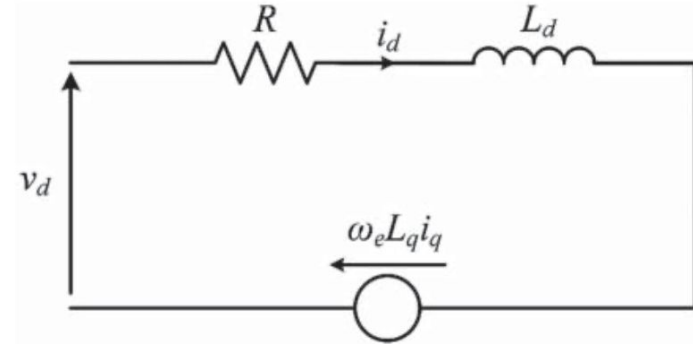
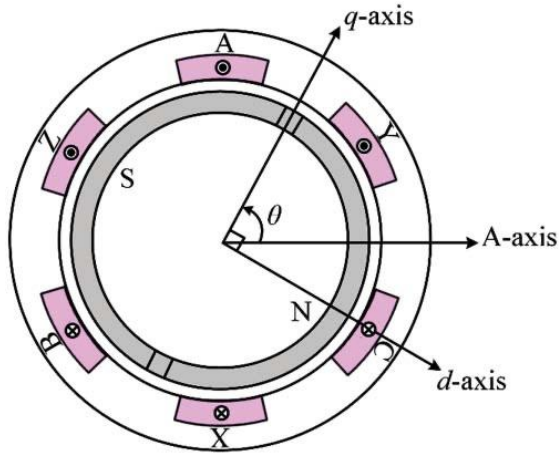
The principle of PM synchronous machine is based on the interaction of **sinusoidal back electromotive force (EMF)** waveforms and **sinusoidal armature (stator) current** waveforms.





PMSM - Modeling

The machine model is expressed about the rotor d-q reference frame.





PMSM - Modeling

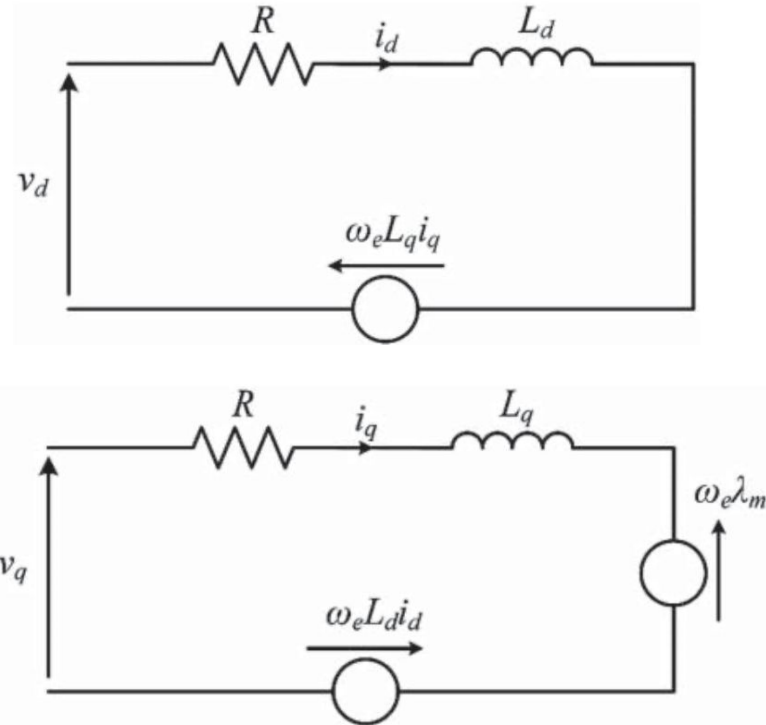
$$v_d = Ri_d + p\lambda_d - \omega_e \lambda_q$$

$$v_q = Ri_q + p\lambda_q + \omega_e \lambda_d$$

Where v_d and v_q are, respectively, the d -axis and q -axis components of stator voltages, i_d and i_q are, respectively, the d -axis and q -axis components of stator currents, and λ_d and λ_q are the d -axis and q -axis components of flux linkages, which can be expressed as:

$$\lambda_d = L_d i_d + \lambda_m$$

$$\lambda_q = L_q i_q$$



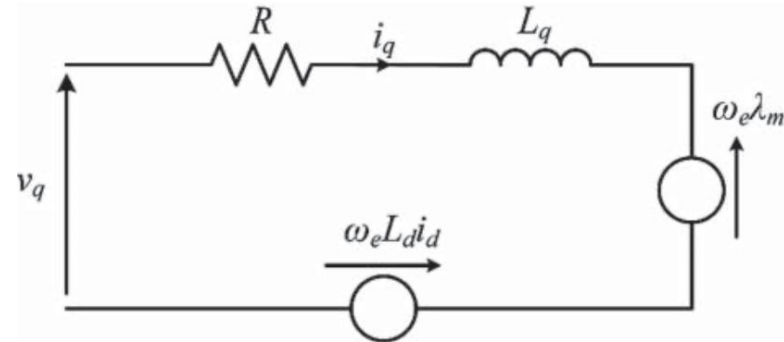
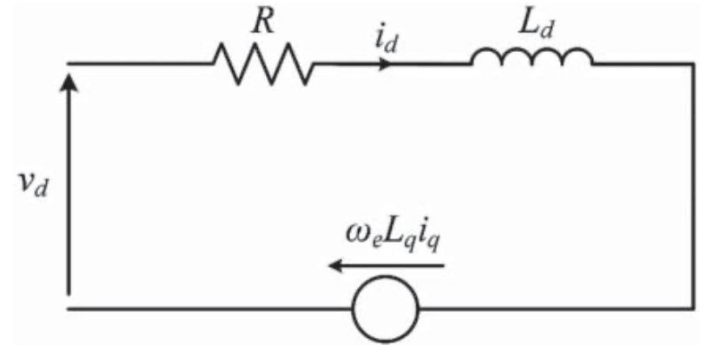


PMSM - Modeling

The electromagnetic developed output torque is equal to:

$$T_e = \frac{3}{2}P(\lambda_m i_q + (L_d - L_q)i_d i_q)$$

The developed torque is composed of two torque components. The first component corresponds to the **reaction torque** between the PM flux and q -axis armature current, while the second component corresponds to the **reluctance torque** due to the difference between d -axis and q -axis inductances.

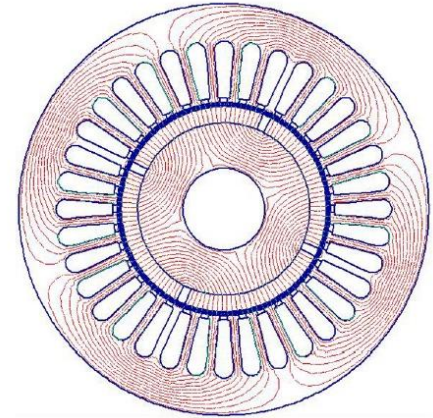




BLDC – Working principle

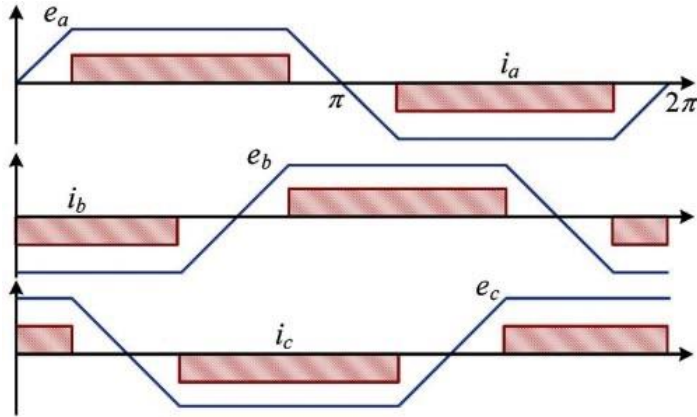
The principle of a BLDC machine is based on the interaction of **trapezoidal back EMF waveforms** and **rectangular armature current waveforms**.

BLDC motors are also referred to as **electronically commutated motor**. As with mechanical commutation, the electronic commutation helps in achieving unidirectional torque similar to a conventional DC machine.





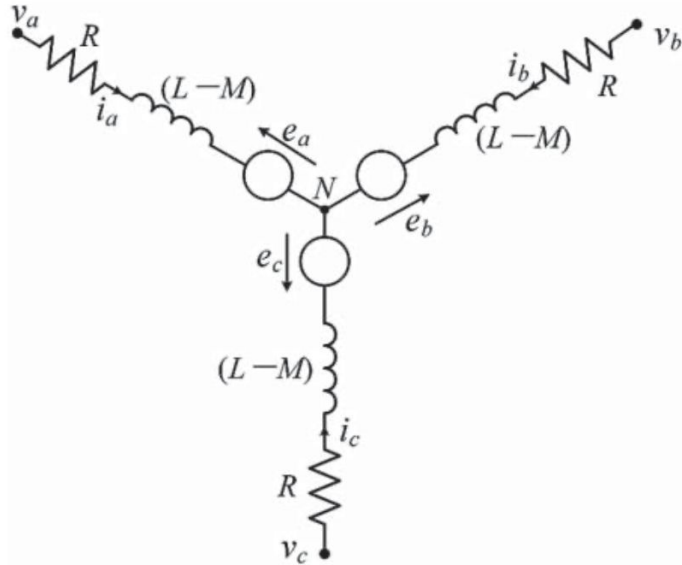
BLDC - Modeling



In a BLDC machine the back EMF waveforms are trapezoidal, rather than sinusoidal. Thus, **the d - q modeling** technique developed for the PM synchronous machine **is not applicable**. Since the flux linkage is nonsinusoidal, it is necessary to derive the model of the PM brushless machine in terms of its **state variables**.



BLDC - Modeling

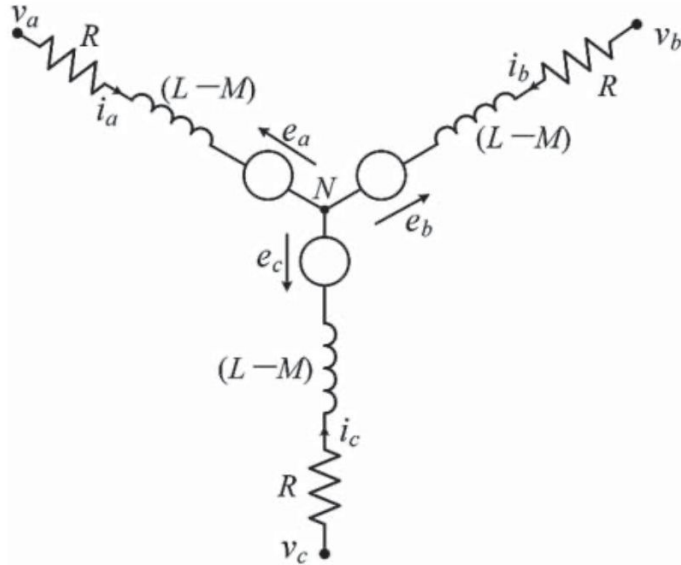


$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + p \begin{bmatrix} L_{aa} & L_{ab} & L_{ac} \\ L_{ba} & L_{bb} & L_{bc} \\ L_{ca} & L_{cb} & L_{cc} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix}$$

- v_a , v_b , and v_c are the instantaneous three-phase stator voltages;
- i_a , i_b , and i_c are the instantaneous three-phase stator currents;
- e_a , e_b , and e_c are the instantaneous three-phase back EMFs;
- R is the armature resistance;
- L_{ij} ($i = a, b, c$ and $j = a, b, c$) are the mutual inductances which are symmetrical.



BLDC - Modeling



The developed torque T_e is the output power divided by the mechanical speed, which is given by:

$$T_e = \frac{P_e}{\omega_r} = P \frac{(e_a i_a + e_b i_b + e_c i_c)}{\omega_e}$$

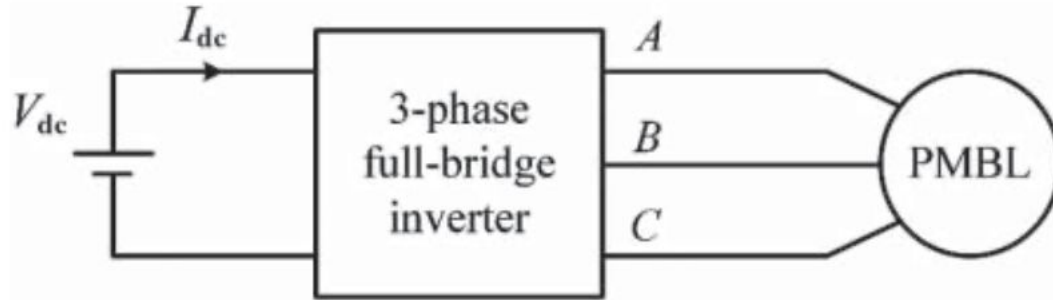
As each phase back EMF is a **bipolar trapezoidal waveform** displaced by 120° (electrical), each current waveform needs to be 120° on positive and 120° on negative within a period to ensure the synchronism and hence produce the desired steady torque without torque pulsations.



Inverters for PM Brushless Motors



PM Brushless Drive



While the PM synchronous motor and PM BLDC motor are fed by sinusoidal current and rectangular current, respectively, **the corresponding inverter topologies are essentially the same** – a three-phase full-bridge inverter. Nevertheless, **their switching schemes are different**:

- the pulse-width modulation (PWM) control for BLAC operation of the PM synchronous motor;
- the stepwise control for BLDC operation of the PM BLDC motor.



Inverter requirements

EV propulsion requires the motor drives offering **four-quadrant characteristics** for:

- forward motoring;
- forward regeneration;
- backward motoring;
- backward regeneration.

Particularly, the forward regeneration is essential to **capture the braking energy** which in turn charges the EV battery, leading to increase the driving range per charge by over 10%.

	Quadrant	Torque	Speed
Forward motoring	I	Positive	Positive
Backward regeneration	II	Positive	Negative
Backward motoring	III	Negative	Negative
Forward regeneration	IV	Negative	Positive

	Quadrant	DC current	AC sequence
Forward motoring	I	Positive	Positive
Backward regeneration	II	Negative	Negative
Backward motoring	III	Positive	Negative
Forward regeneration	IV	Negative	Positive



PM Brushless Drive

Quadrant I: forward motoring

the torque and speed are positive so that the input and output powers of the inverter are positive. Thus, the average input DC current to the inverter has to be positive since the battery voltage is naturally positive. As the motor speed is positive, the phase sequence of the stator terminals is positive, namely, A-B-C.

Quadrant IV: forward regeneration

the speed is positive but the torque is negative so that the input and output powers of the inverter are negative. Thus, the average input DC current becomes negative so that the braking energy is converted to charge the battery. As the motor speed is still positive, the stator phase sequence is positive.

The operations at **Quadrants III and II** are similar to that at Quadrants I and IV, respectively, except that the **motor speed is negative** and the corresponding stator phase sequence is negative, namely, A-C-B.



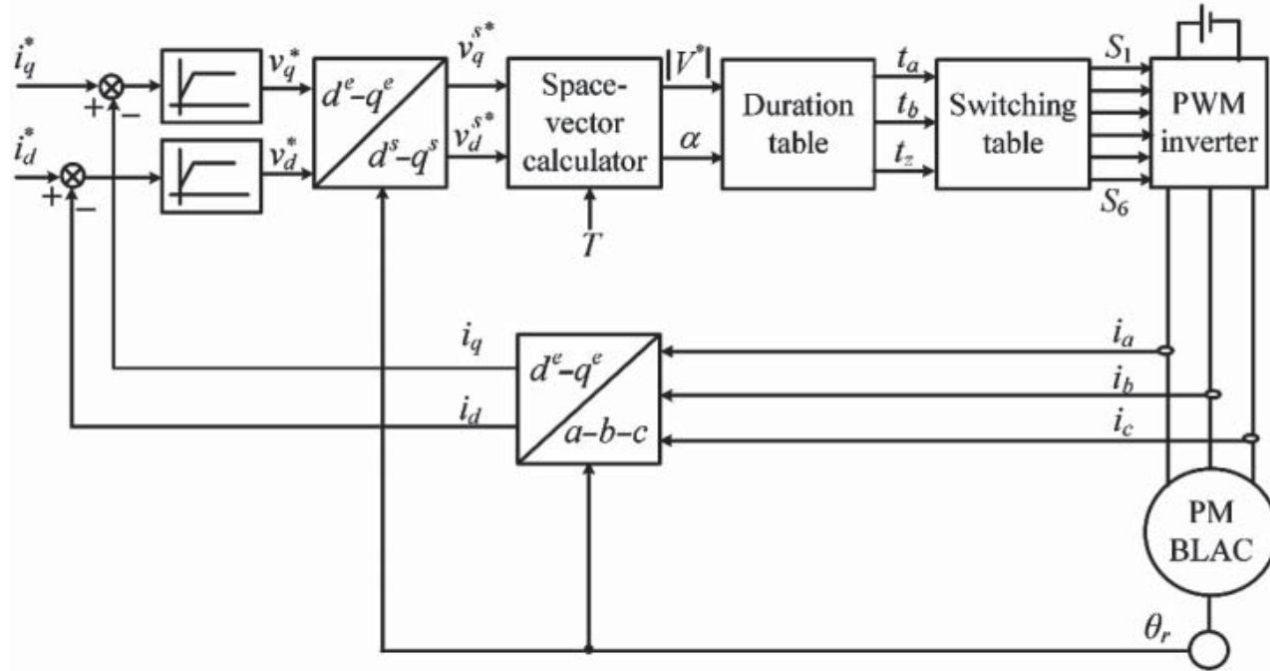
Switching Schemes for Brushless AC Operation

The power inverter topology used for the PM synchronous motor drive operating at the BLAC mode for EV propulsion is essentially the same as that for the induction motor drive. That is, the preferred power inverter is based on the **voltage-fed full-bridge** topology. As the inverter has to act on the commands of magnitude, frequency, and phase of the output voltages and currents, there are various switching schemes to achieve such control.

The most diffused one is **Space Vector Modulation** (SVM).



Space Vector Modulation





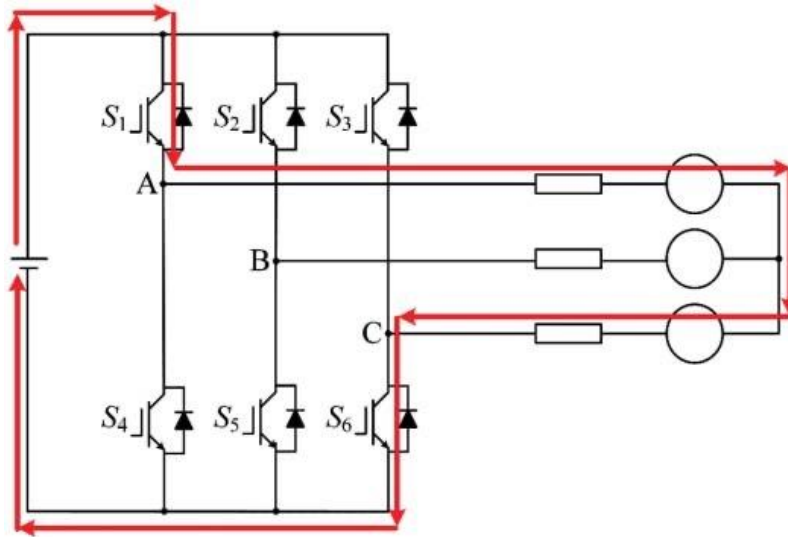
Switching Schemes for Brushless DC Operation

The PM BLDC machine is driven by strokes in accordance with the rotor position. These strokes must be properly applied to the active phases of the three-phase armature winding so that the angle between the stator flux and the rotor flux is kept close to 90° to develop **the maximum torque**. There are two kinds of switching schemes to drive this PM BLDC machine:

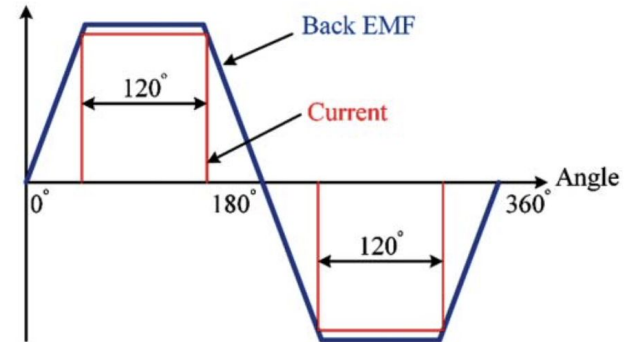
- Two-phase 120° conduction scheme;
- Three-phase 180° conduction scheme.



Switching Schemes for Brushless DC Operation



Intervals (°)	Devices on	Phases A, B, C
0–60	S_1, S_6	+, 0, –
60–120	S_2, S_6	0, +, –
120–180	S_2, S_4	–, +, 0
180–240	S_3, S_4	–, 0, +
240–300	S_3, S_5	0, –, +
300–360	S_1, S_5	+, –, 0





PM Brushless Motor Control

Since the operating waveforms of the PM synchronous and PM BLDC machines are different, the corresponding motor control strategies are naturally different.

Essentially, the PM synchronous motor can adopt those control strategies that have been developed for the **induction motor** because both types of motors are based on sinusoidal waveforms, whereas the PM brushless motor needs to adopt **dedicated control strategies** because of its nonsinusoidal operating waveforms.



PMSM Motor Control

The PM synchronous motor can adopt those sophisticated control strategies that have been developed for the induction motor, such as the FOC and direct torque control. Between them, **the FOC has been widely applied** to the PM synchronous motor for EV propulsion.

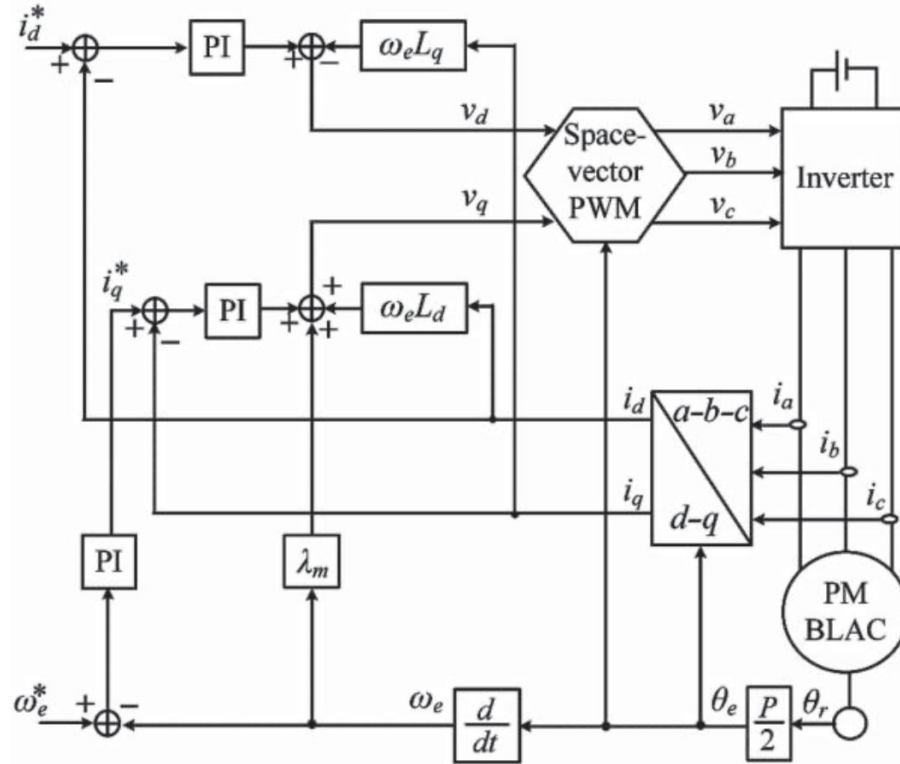
Meanwhile, differing from the induction motor, the PM synchronous motor adopts PM field excitation which is inherently uncontrollable.

In order to provide the constant-power operation for EV cruising, the **flux-weakening control** of the PM synchronous motor is highly desirable.

On the other hand, in order to get rid of the costly position encoder, **the position sensorless control** has been actively developed for the PM synchronous motor.



PMSM – Field Oriented Control





BLDC Motor Control

As the PM BLDC motor is driven when the stator flux and the rotor flux is kept close to 90° , it can readily produce the maximum torque per ampere in the region of constant-torque operation. This enables the PM BLDC motor taking a definite advantage over the PM synchronous motor which relies on using the d–q coordinate transformation.

However, in the absence of d–q coordinate transformation or FOC, **the control scheme** for constant-power operation of the PM BLDC motor **is much more complicated** than that of the PM synchronous motor.

Thus, the **phase-advance angle control** can enable the PM BLDC motor offering constant-power operation for EV cruising. Meanwhile, the **position sensorless control** has been actively developed for the PM BLDC motor.



Conclusions



Lessons outcome

Among the three main types of commercially available EV motor drives, namely the DC, induction, and PM brushless, the PM brushless motor drives take the definite advantages of **higher efficiency, higher power density, and higher torque density**, which are highly desirable for EV propulsion. Currently, there is no doubt that the PM brushless motor drives are **the preferred technology** for EVs.

Between the two members of PM brushless motor drives, the PM synchronous motor drive is **more mature** than the PM BLDC motor drive while they have their individual merits. That is, the PM synchronous motor drive has relatively **higher efficiency** and **lower torque ripple**, whereas the PM BLDC motor drive has relatively **higher power density** and **higher torque density**.

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